PAPER_481 — 18-System UQFF Module Suite: F_U_Bi_i C++ Implementations for Astrophysical Systems (Oct 2025 Batch)

Abstract

This paper documents the complete C++ module implementation suite for 18 astrophysical systems encoded in the UQFF framework (Unified Quantum Field Framework). Each module encapsulates the full $F_{U_{Bi_i}}$ Master Unified Field Equation with system-specific parameters stored in a `std::map<std::string, std::complex<double>>` dictionary, enabling runtime parameter updates, dynamic sub-term computation, and descriptive equation output. This batch extends the existing individual-system module collection (Abell 2256 v1, Centaurus A) to a comprehensive library covering TDEs, compact clusters, pulsars, interacting galaxies, AGN jets, symbiotic binaries, solar-system aurorae, and star-forming regions.

Source: `grok_share_bdfb3a05b06.txt` (~11,592 lines), Grok analysis of 18 × `.docx` attachments (Sept–Oct 2025), Session 126 extraction.

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1. Unified Field Architecture

All 18 modules share the same computational skeleton:

$$F_{U_{Bi_i}} \approx \left(\int_0^{x_2} \mathcal{I}(r,t)\, dx \right) \approx \mathcal{I} \cdot x_2$$

where the integrand $\mathcal{I}$ sums all force contributions:

$$\mathcal{I} = -F_0 + \frac{m_e c^2}{r^2} \mathrm{DPM}_{mom} \cos\theta + \frac{GM}{r^2} \mathrm{DPM}_{grav} + \rho_{vac,UA} \mathrm{DPM}_{stab} + k_{LENR} \left(\frac{\omega_{LENR}}{\omega_0} \right)^2 + k_{act} \cos(\omega_{act} t + \phi) + k_{DE} L_X$$

with quadratic root approximation $x_2 \approx -1.35 \times 10^{172}$ (universal constant).

Sub-equation API (identical for all modules): - `computeF(t)` — full $F_{U_{Bi_i}}$ (integral approx) - `computeCompressed(t)` — integrand $\mathcal{I}$ - `computeResonant()` — DPM magnetic resonance - `computeBuoyancy()` — $U_{b1} = \beta_i V_{infl,UA} \rho_{vac,A} a_{\mathrm{univ}}$ - `computeSuperconductive(t)` — $U_i = \lambda_i \frac{\rho_{vac,SCm}}{\rho_{vac,UA}} \omega_s \cos\left(\frac{t}{t_{scale}}\right)(1 + f_{TRZ})$ - `computeCompressedG(t)` — gravity analog $g(r,t)$ - `getEquationText()` — LaTeX-formatted description - `updateVariable` / `addToVariable` / `subtractFromVariable` — runtime parameter management

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2. System Parameter Catalogue

2.1 Universal Constants (All Modules)

Constant	Value	Units
G	6.6743×10^{-11}	m^3 kg ^{1} s ^{2}
c	3×10^{8}	m/s
$\hbar$	1.0546×10^{-34}	J $\cdot$ s
q	1.6×10^{-19}	C
m_e	9.11×10^{-31}	kg
μ_B	9.274×10^{-24}	J/T
$\rho_{vac,UA}$	7.09×10^{-36}	kg/m^3
k_{LENR}	10^{-10}	—
ω_{LENR}	$2\pi \times 1.25 \times 10^{12}$	rad/s
F_0	1.83×10^{71}	N
x_2	-1.35×10^{172}	—
F_ν	9.07×10^{-42}	N
E_{cm} (LEP ref)	3.0264×10^{-8}	J (189 GeV)

2.2 System-Specific Parameters

Module	Object Type	M (kg)	r (m)	ω_0 (rad/s)	L_X (W)	t default (s)	$E_{cm,astro}$ (J)
ASASSN14liUQFFModule	TDE	1.989×10^{37}	3.09×10^{18}	10^{-12}	10^{37}	9.504×10^{6}	1.24×10^{24}
CrabNebulaUQFFModule	PWN/SNR	10^{31}	4.73×10^{16}	10^{-12}	10^{27}	3.06×10^{10}	1.24×10^{24}
ElGordoUQFFModule	Galaxy Cluster	4.97×10^{45}	3.09×10^{22}	10^{-15}	2×10^{38}	2.21×10^{16}	1.24×10^{24}
ESO137UQFFModule	Jellyfish Galaxy	2×10^{41}	6.17×10^{21}	10^{-15}	10^{34}	7.72×10^{14}	1.24×10^{24}
IC2163UQFFModule	Interacting Galaxy	1.989×10^{40}	3.09×10^{20}	10^{-12}	10^{37}	1.26×10^{15}	1.24×10^{24}
J1610UQFFModule	High-z Quasar	1.73×10^{40}	9.63×10^{20}	10^{-15}	10^{39}	3.156×10^{14}	1.24×10^{24}
JupiterAuroraeUQFFModule	Planetary Aurorae	1.898×10^{27}	7.1492×10^{7}	10^{-12}	10^{26}	60.0	1.24×10^{24}
LagoonNebulaUQFFModule	H II Region	10^{36}	2.36×10^{17}	10^{-12}	10^{32}	10^{13}	1.24×10^{24}
M87JetUQFFModule	AGN Relativistic Jet	1.29×10^{40}	4.63×10^{19}	10^{-15}	10^{34}	3.156×10^{14}	1.24×10^{24}
NGC1365UQFFModule	Barred Spiral AGN	7.17×10^{41}	9.46×10^{20}	10^{-15}	10^{36}	1.1×10^{16}	1.24×10^{24}
NGC2207UQFFModule	Interacting Galaxy	3.978×10^{40}	4.40×10^{20}	10^{-12}	10^{37}	1.26×10^{15}	1.24×10^{24}
RAquariiUQFFModule	Symbiotic Binary	3.978×10^{30}	2.18×10^{15}	10^{-12}	10^{32}	1.4×10^{9}	1.24×10^{24}

Module	Object Type	M (kg)	r (m)	ω_0 (rad/s)	L_X (W)	t default (s)	$E_{cm,astro}$ (J)
`SgrAStarUQFFModule`	SMBH (Milky Way)	8.56×10^{36}	6.17×10^{18}	10^{-15}	10^{36}	10^{15}	1.24×10^{24}
`SPTCLJ2215UQFFModule`	Cool-Core Cluster	1.46×10^{45}	3.09×10^{22}	10^{-15}	2×10^{38}	2.21×10^{16}	1.24×10^{24}
`StephanQuintetUQFFModule`	Compact Group	2×10^{39}	3.09×10^{22}	10^{-15}	10^{38}	10^{16}	1.24×10^{24}
`VelaPulsarUQFFModule`	Pulsar/PWN	2.8×10^{30}	1.7×10^{17}	10^{-12}	10^{27}	3.47×10^{11}	1.24×10^{24}

Notes: - All modules share universal DPM parameters: $\rho_{vac,UA} = (7.09 \times 10^{-36} + 10^{-37} i)$ kg/m³, $\text{DPM}_{mom} = 0.93 + 0.05i$, $\text{DPM}_{grav} = 1.0 + 0.1i$ - $E_{cm} = 3.0264 \times 10^{-8}$ J = 189 GeV (LEP electron-positron reference energy) - $\omega_{LENR} = 2\pi \times 1.25 \times 10^{12}$ rad/s (LENR THz resonance, universal)

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3. Notable Physical Observations

3.1 Jupiter Aurorae — UQFF at Planetary Scale

The `JupiterAuroraeUQFFModule` is uniquely significant: it applies the $F_{U_{Bi_i}}$ framework to a **planetary-scale** object, with $M = 1.898 \times 10^{27}$ kg (Jupiter mass) and $r = 7.1492 \times 10^7$ m (Jupiter equatorial radius). Default $t = 60$ s captures auroral emission timescale. This demonstrates UQFF universality across 18 orders of magnitude in mass (Jupiter M to galaxy cluster M).

3.2 R Aquarii — Symbiotic Binary on HST-Observable Timescale

$t = 1.4 \times 10^9$ s = 44.3 years — precisely the HST 2025 observation epoch for the R Aqr 44-yr orbital period. $\omega_0 = 10^{-12}$ rad/s reflects the binary orbital resonance frequency.

3.3 Universal DPM Resonance Formula

$$\text{DPM}_{res} = \frac{g_L \mu_B B_0}{\hbar \omega_0}$$

For $B_0 = 10^{-9}$ T (galaxy cluster field), $\omega_0 = 10^{-15}$ rad/s: $\text{DPM}_{res} \approx 1.76 \times 10^{17}$. For $B_0 = 10^{-5}$ T (TDE field), $\omega_0 = 10^{-12}$ rad/s: $\text{DPM}_{res} \approx 1.76 \times 10^{15}$.

3.4 LENR Dominance Signature

At low ω_0 (cluster scale, 10^{-15} rad/s):

$$k_{LENR} \left(\frac{\omega_{LENR}}{\omega_0} \right)^2 = 10^{-10} \times (2\pi \times 1.25 \times 10^{12} / 10^{-15})^2 \approx 10^{-10} \times 6.2 \times 10^{54} \approx 6.2 \times 10^{44}$$

This LENR term dominates all other integrand contributions for cluster-scale systems.

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4. Module Architecture Notes

4.1 File Inventory (Created Session 126)

Header File	Impl File	Size (h/cpp, chars)
`ASASSN14liUQFFModule.h`	`ASASSN14liUQFFModule.cpp`	2,748 / 13,968
`CrabNebulaUQFFModule.h`	`CrabNebulaUQFFModule.cpp`	2,740 / 13,967
`ElGordoUQFFModule.h`	`ElGordoUQFFModule.cpp`	2,728 / 13,883
`ESO137UQFFModule.h`	`ESO137UQFFModule.cpp`	2,706 / 13,925
`IC2163UQFFModule.h`	`IC2163UQFFModule.cpp`	2,696 / 13,835
`J1610UQFFModule.h`	`J1610UQFFModule.cpp`	2,694 / 13,818
`JupiterAuroraeUQFFModule.h`	`JupiterAuroraeUQFFModule.cpp`	2,783 / 14,149
`LagoonNebulaUQFFModule.h`	`LagoonNebulaUQFFModule.cpp`	2,761 / 14,032
`M87JetUQFFModule.h`	`M87JetUQFFModule.cpp`	2,694 / 13,900
`NGC1365UQFFModule.h`	`NGC1365UQFFModule.cpp`	2,715 / 13,911
`NGC2207UQFFModule.h`	`NGC2207UQFFModule.cpp`	2,709 / 13,902
`RAquariiUQFFModule.h`	`RAquariiUQFFModule.cpp`	2,726 / 13,928
`SgrAStarUQFFModule.h`	`SgrAStarUQFFModule.cpp`	2,722 / 13,927
`SPTCLJ2215UQFFModule.h`	`SPTCLJ2215UQFFModule.cpp`	2,766 / 14,035
`StephanQuintetUQFFModule.h`	`StephanQuintetUQFFModule.cpp`	2,799 / 14,125
`VelaPulsarUQFFModule.h`	`VelaPulsarUQFFModule.cpp`	2,756 / 14,004
`StarMagicUQFFModule.h`	*(cpp existed)*	2,994 / —

4.2 Pre-existing Modules (Skipped)

- `Abell2256UQFFModule.h/.cpp` — existed from PAPER_472
- `CentaurusAUQFFModule.h/.cpp` — existed from PAPER_480
- `SMBHUQFFModule.h/.cpp` — existed from PAPER_468/470
- `UQFFBuoyancyModule.h/.cpp` — existed from PAPER_479
- `StarMagicUQFFModule.cpp` — existed from PAPER_144 (header added)

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5. Integration Pathway

Phase A: MAIN_1 Integration

Register all 18 modules in `MAIN_1_CoAnQi.cpp` under `SOURCE_SESSION126_MODULES` namespace. Each module contributes to the physics term registry with: - `computeF(t)` → `F_U_Bi_i` value - `computeCompressedG(t)` → gravitational analog $g(r, t)$

Phase B: CP2 Calculator

Add `IndividualSystemUQFF18Calculator` to `CondensedPhysics2.py` wrapping all 18 `computeF()` results in a unified dataset response. Target: CP2 class count 602 → 603.

Phase C: CP4 Registry

Add `Session126GrokShareBdfb3a05b06HubCalculator` as CP4 entry #105.

6. Validation Cross-References

Module	Existing Paper	New Module Files
ASASSN14li	PAPER_351	ASASSN14liUQFFModule.h/.cpp
Crab Nebula	PAPER_220, 256, 290–292	CrabNebulaUQFFModule.h/.cpp
El Gordo	PAPER_350	ElGordoUQFFModule.h/.cpp
ESO 137-001	PAPER_338 (catalogue)	ESO137UQFFModule.h/.cpp
IC 2163	PAPER_338 (catalogue)	IC2163UQFFModule.h/.cpp
J1610+1811	PAPER_161, 360	J1610UQFFModule.h/.cpp
Jupiter Aurorae	PAPER_157, 338	JupiterAuroraeUQFFModule.h/.cpp
Lagoon Nebula	PAPER_305–307	LagoonNebulaUQFFModule.h/.cpp
M87 Jet	PAPER_093, 346	M87JetUQFFModule.h/.cpp
NGC 1365	PAPER_338 (catalogue)	NGC1365UQFFModule.h/.cpp
NGC 2207	PAPER_338 (catalogue)	NGC2207UQFFModule.h/.cpp
R Aquarii	PAPER_352	RAquariiUQFFModule.h/.cpp
Sgr A*	PAPER_067, 149, 234, 366	SgrAStarUQFFModule.h/.cpp
SPT-CL J2215	PAPER_349	SPTCLJ2215UQFFModule.h/.cpp
Stephan's Quintet	PAPER_348	StephanQuintetUQFFModule.h/.cpp
Vela Pulsar	PAPER_337, 066	VelaPulsarUQFFModule.h/.cpp
